

FlexPhys: A Workshop Cookbook for Operationalizing Data Physicalization Research Questions

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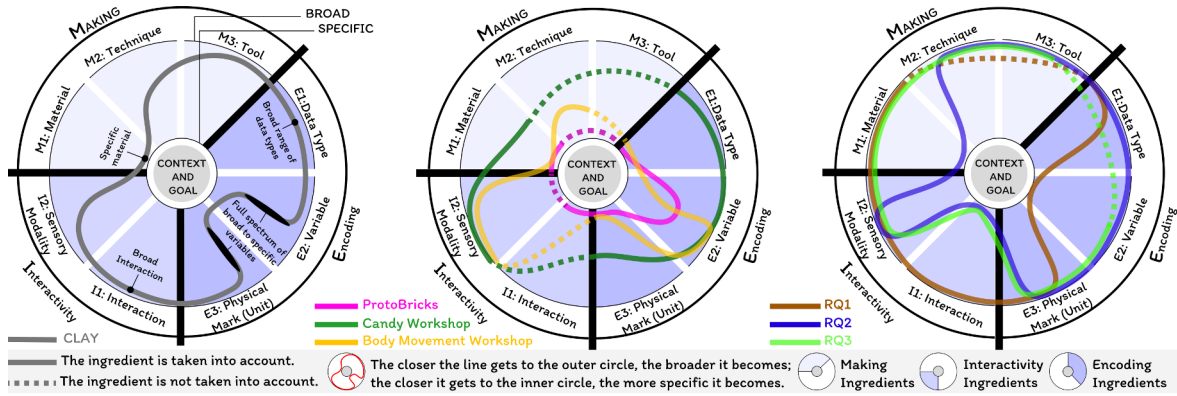


Figure 1: The FlexPhys cookbook. Researchers can draw a path passing through the eight ingredients of the cookbook. The core ingredient is the context and the goal. Left: The light grey line shows the ingredients covered by a workshop study (CLAY workshop); Middle: Three physicalization workshop examples [7, 18, 25] described using FlexPhys; Right: Three workshop scenarios that answer three types of research questions generated using FlexPhys.

ABSTRACT

We introduce FlexPhys, a cookbook that researchers can use to operationalize data physicalization research questions through workshop design. While guidelines exist for running workshops in educational contexts, designing a data physicalization workshop when the goal is to answer research questions is an ad-hoc process for which little guidance exists, but for which many choices must be made (e.g., in terms of materials, tools, and data). We draw from our experience designing data physicalization workshops and from reviewing three existing workshops, to distill the cookbook's core ingredient (context and goal) and eight additional ingredients related to making (material, tool, technique), data encoding (data type, variable, mark/unit), and interactivity (interaction, sensory modality). We then show how FlexPhys can be used to describe and compare physicalization workshops, and to generate workshops that address specific data physicalization research questions.

Index Terms: data physicalization, methods, workshops.

1 INTRODUCTION

Data physicalization (or simply physicalization), in which data is encoded through material properties [14], has seen increased interest from the visualization community in the past decade. Within this space, two types of studies leveraging workshops have been conducted by researchers. The first type discussed in the literature has educational purposes—often to familiarize non-experts with visualization concepts (e.g., [11, 12, 21]). The second type addresses

data physicalization research questions—for example, by investigating which physical variables people might use in which context, with which material, and through which sensory modality and interaction [25, 20, 18]. We focus on this second category.

Translating research questions into a specific workshop design requires some form of operationalization. For example, Huron et al. made the choice of using off-the-shelf materials like wooden tiles or craft materials to familiarize novices with visualization concepts [12]. Similarly Zizzi et al. used dancing as a medium to explore potential data that could be physicalized in body movement [25]. By making these choices, researchers operationalize research questions by defining precisely the materials, variables, and measurement methods they will leverage in their study.

Compared to on-screen visualization, data physicalizations involve an important material aspect. The material properties, the form factors, and the affordances of the physical objects, provide unique opportunities for incorporating sensory modalities and tangible interactions that lack in visualization contexts. Compared to visualization, no guidelines currently support the operationalization of research questions to investigate the many facets of data physicalization. Researchers have proposed guidelines and frameworks that describe how and when to organize (non-physical) visualization workshops [5, 16, 17], but the tangible nature of physicalization raises many questions related to hands-on participation and manual handling of materials. While some data physicalization researchers have reflected on their processes and experiences [11, 12] to provide guidelines, those guidelines are specific to particular research questions and do not constitute a systematic and flexible framework that others can use to design their workshops.

We propose a generic workshop cookbook that guides researchers in the operationalization of their research questions, in the context of data physicalization. We arrived at this cookbook by combining our own experience running a series of physicalization workshops with clay, with an analysis of three existing workshop instructions [11, 12, 24] from the literature. Then, we demonstrate the usefulness of FlexPhys to describe and compare three additional

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data physicalization workshops [7, 18, 25]. Further, we show how researchers can use FlexPhys to generate workshop designs by systematically considering all ingredients of the cookbook.

We make two contributions in this paper:

1. We propose FlexPhys, a generic cookbook that researchers can use to design data physicalization workshops. FlexPhys serves as an initial framework to operationalize research questions through physicalization workshop design. It provides one core ingredient (context and goal of the physicalizations) and eight additional ingredients for operationalizing research questions, by considering making (material, tool, technique), encoding (data type, variable, mark/unit), and interactivity (interaction, sensory modality).
2. We demonstrate the descriptive-comparative power of FlexPhys by showing how it can help describe and compare existing workshops; we then demonstrate the generative power of FlexPhys by presenting three scenarios of possible workshop structures that could be implemented to answer different kinds of data physicalization research questions.

2 METHODOLOGY AND PROCESS

Our experience conducting workshops to study data physicalization motivated the creation of a cookbook to operationalize research questions. Unlike lab experiments which aim for control and reproducibility, workshop studies lean towards realism. Such workshops are close to field experiments or studies in Runkel and McGrath's circumplex of research strategies [22], they tend to be open-ended and challenging to structure. While studying the impact of freeform materials on data physicalization design, we iterated through multiple workshop designs, corresponding pilot studies, and debrief sessions, before we could identify the elements we ought to consider in the actual workshop. For instance, what kinds of materials we should use, what range of tools we should provide in the workshop, if we should explicitly ask participants to represent data to be perceived using one sensory modality like tactile and so on.

FlexPhys emerged from combining our experience designing this workshop (that we refer to as **CLAY**) and an analysis of three data physicalization workshop instructions that researchers thoroughly described and reflected upon: **VISKIT** [11], Let's Get Physical [12] (that we refer to as **LGP**), and a constructive classroom exercise [24] (that we refer to as **CCE**).

VISKIT is a manual for running constructive visualization workshops to familiarize lay people with visualization authoring concepts. It uses tangible tiles for physicalization and provides six high level steps: presenting **VISKIT**, authoring, reading, editing, critiquing, and conclusion [11]. **LGP** aims to facilitate engagement in physicalization activities by introducing sets of activities, scenarios, and data cards, a set of craft materials with different properties, and documentation tools [12]. **CCE** is a version of **VISKIT** adapted to a classroom environment for teaching information visualization. The authors expanded the workshop material and included Lego blocks and clay in addition to the pre-made tiles, with the aim to enhance creativity and increase the variety of the physicalizations. They also adjusted the steps to: construction, self-evaluation, communication, and peer evaluation [24].

Our analysis of workshops from the literature focused on what participants were given, what they could do, and the constraints or freedom they were given in the instructions. We reviewed materials, interactions, the data involved, the physical variables and physical marks. Some of these 'ingredients' were explicitly part of the workshop instructions (e.g., using a specific material such as wooden tiles or Lego blocks) while others were implicitly incorporated and broadly explored by participants (e.g., how to interact with wooden tiles or which variables to use for data representation).

In **CLAY** our goal was to investigate freeform physicalization and compare it to token-based physicalization. The **CLAY** work-

shop consisted of three iterations of physicalization activities in which we provided participants with a step by step guide that included a dataset about physical activity, mood, and fatigue over a week, and a list of visual and haptic variables [10] to choose from. Participants had to experiment with two physicalization approaches: constructive and sculptural. We provided participants with a wide range of clay work tools, paint, wire armature, and clay as the main material. We also introduced participants to three clay work techniques (coiling, slabbing, and pinching). Using a documentation handbook we encouraged participants to document and reflect on aspects such as data mapping and interaction.

By analyzing the different components affecting data physicalization in the **CLAY** workshop, we added new components consisting of technique, tool, and sensory modality to the components identified from the literature. Derived from the insights from the above-mentioned sources, we propose FlexPhys, a cookbook for operationalizing data physicalization workshops.

3 FLEXPHYS INGREDIENTS

In this section, we introduce a core ingredient and the other eight ingredients that make up the FlexPhys cookbook for designing workshops. Figure 1 shows the FlexPhys cookbook applied to our cases. We explain each ingredient and illustrate it using the four workshops we introduced earlier (**CLAY**, **VISKIT**, **LGP**, and **CCE**).

► **CONTEXT AND GOAL** – Refers to the context in which the workshop activities unfold, and the goals of the workshops in terms of physicalization creation. This ingredient is necessary to any data physicalization workshop. Context and goal can be defined either by workshop organizers or by workshop participants. In **VISKIT**, the context and goal are defined by organizers: participants are asked to imagine they are trying to help a friend who is having trouble with expenses. In **LGP**, participants choose their own physicalization goals from a set of activity and scenario cards. In **CLAY**, one participant created physicalizations in the context of personal informatics and explained their goal was to “*sensibilize the user to his feelings so he can choose which activity is better for him*”; other participants reflected on the importance of “*defining a use case*” to help them stay focused and produce a meaningful physicalization.

The other eight ingredients of the cookbook should be taken into account based on the research questions. For instance, one might be interested in the effect of materials in the data physicalization process while others may not consider it as research focus. We structure these eight ingredients into three categories: making (material, tool, technique), encoding (data type, variable, mark/unit) and interactivity (interaction, sensory modality). There are two concentric circles on the wheel of the cookbook that show two different states: Specific and Broad. Specific means that researchers have a particular interest regarding an ingredient, for instance a specific material like clay or a specific technique like laser cutting. Broad means that researchers have an exploratory approach regarding an ingredient, for instance considering a wide range of craft materials.

3.1 Making Ingredients

Making refers to bringing data representations into existence in the physical world—a process that is also called ‘rendering’ [6]. We identified three making ingredients: material, technique, and tool.

► **M1: MATERIAL** – Refers to “physical means to encode the data” [12]. M1 has been used in various ways and for different purposes in data physicalization. For example, M1 is specific in **VISKIT** as restricted to wooden tiles, to enable novice authoring of data visualizations. Conversely, M1 is broad in **LGP** as they provide a broad range of craft materials with a wide range of properties such as rigidity, elasticity and malleability. By keeping M1 broad, the workshop organizers did not limit participants’ creativity based

on constrained social or technological choices, and allowed participants to select material they would be familiar with and comfortable working with, which avoided the need for an instruction phase.

► **M2: TECHNIQUE** – Refers to specific ways of working with a material. For instance, when working with clay, people employ techniques like coiling and pinching, and in working with textile, they employ techniques like knitting and weaving. Compared to materials, techniques (and tools) are underexplored in the physicalization community (although exceptions exist, e.g., [25, 15]).

M2 is irrelevant in VISKIT, LGP, and CCE that either rely on pre-made, fixed units like wooden tiles, or that provide malleable material like clay but with technique not being considered. In CLAY, we observed that introducing clay work techniques (pinching, coiling, and slabbing) in the data physicalization process opens up new opportunities and unique data representation. For example, one participant represented tiredness with the metaphor of a flower using the pinching technique.

► **M3: TOOL** – Refers to hardware and software that participants are provided with to make physicalizations. In LGP, a broad range of tools are provided, including ruler, scissors, cutter, pencil, and glue, to facilitate the making process by providing affordances such as cutting, attaching, and annotating. In CLAY, we provided various clay tools and we observed that the affordances of tools play an important role in participants' expressions of ideas. For example, one participant explained that "[this object] was supposed to be some kind of fire, but how do you sculpt fire with clay? There are other shapes that were easier to sculpt. Well, obviously the poop was easier [...] with the extruder [...]" but then noted in their documentation handbook that in another iteration they "tried different tools to render different kinds of [snake] scales". Although not employed in the workshops we reviewed, tools can include digital fabrication tools such as additive tools (e.g., 3D printers) and subtractive tools (e.g., laser cutter) [6], as mentioned in LGP.

3.2 Data Mapping Ingredients

Similar to how in visualization, data mapping happens through assigning data attributes to the variables of visual marks [3], in physicalization, data mapping happens through assigning data attributes to the properties or geometries of the material [14]. We identified three data mapping ingredients: data type, variable, and mark/unit.

► **E1: DATA TYPE** – Refers to the properties of the data to be physicalized and its attributes. Attributes can for example be quantitative, ordinal, and nominal [3]. The dataset can be of many different types (e.g., tabular, temporal, multi-dimensional, network [23]), can be large or small, be static or dynamic, and include uncertain or missing data. The dataset must be carefully chosen so that it is relevant to the research questions.

For example, E1 is specific in LGP and VISKIT that only provide tabular datasets with solely quantitative attributes. In CLAY, we wanted to observe differences in data mapping based on data type, therefore we provided a dataset that included quantitative, nominal, and ordinal data attributes.

► **E2: VARIABLE** – Refers to the properties of the material that can be used to encode data. Although variable selection is a crucial part of data mapping in data visualization, in physicalization workshop formats such as LGP, the mapping is not considered as an explicit step but as an implicit part of the data physicalization process. In VISKIT, the mapping is narrowly defined with each tile representing a base unit of value 25. The downside of this approach is that participants to those workshops end up creating similar physicalizations by using similar variables—variables that are often not physical per se, but standard visual ones (e.g., shape, color, and size [9]). In CLAY, we nudged participants to explore more unique physical variables by providing them with a list of visual and haptic variables such as weight, speed, and viscosity.

► **E3: MARK/UNIT** – Refers to measurable physical units which represent data, extending Bertin's definition of visual variables [3].

E3 is specific in workshops that provide pre-made tokens such as wooden tiles, such as VISKIT. E3 is broad in LGP and CCE as they provide variations of materials (e.g., craft materials) with different form factors for participants to create various marks from. It is even broader in CLAY as we used malleable material, which allows participants to create infinite variations of physical marks.

3.3 Interactivity Ingredients

Studying interactions between people and physicalizations is an important aspect of data physicalization studies [14], and is tightly connected to the context and goal of the physicalizations.

► **I1: INTERACTION** – Interaction is a key aspect of data physicalizations that can support cognitive processes, enable active perception, and improve engagement [14]. I1 is not considered in VISKIT. In LGP, although interaction is not explicitly mentioned, participants are provided with activity cards (e.g., convince, discover, collect, enjoy, and simulate) that resulted in them thinking about interaction and integrating interactions in their physicalizations. In CLAY, we asked explicitly that participants design and describe the possible interactions with their data physicalizations. We collected examples of both active and passive interactions.

► **I2: SENSORY MODALITY** – Interaction happens through one or more of the sensory modalities. While VISKIT, LGP, and CCE do not explicitly consider sensory modality in the workshop structure, we did so in CLAY. In CLAY, we encouraged participants to think about which sensory modality a person could rely on when engaging with the physicalization. We did so by providing lists of both visual and haptic variables, and observed that participants created data physicalization where interaction happens through both vision and touch. In section 4, we describe workshop examples that focus explicitly on exploring new sensory modalities.

4 DESCRIPTIVE AND COMPARATIVE POWER OF FLEXPHYS

We show how FlexPhys provides a vocabulary and common ground to describe and compare data physicalization workshops, by analyzing three workshops [7, 18, 25]. Figure 1-middle provides a visual comparison of these three workshops through our cookbook.

★With ProtoBricks, Dominiak et al. [7] report of a physicalization workshop used to investigate how their rapid prototyping toolkit supported physicalization and creativity. The workshop explored the toolkit use in relation to **E2**: the materials used to encode data. The lego blocks used offer several mapping variables (e.g., position, length and color) making it a broad ingredient. Most other ingredients had a narrow scope: **M1** consisted of conductive Lego blocks, **E2** of a dataset on electrical energy consumption that only included quantitative data, **E3** of Lego blocks, and **I1** of assembling and disassembling the blocks. **M2**, **M3** and **I2** were not considered.

★In their Candy workshop, Lallemand et al. [18] studied sensory modalities other than sight and touch in constructive data physicalization. An analysis with FlexPhys places this workshop on the open-ended side of the spectrum. Since sensory modality was an important element of their exploratory study, they kept **I2** broad and asked participants to consider additional sensory modalities in their physicalization. They provided various quantitative, nominal, and ordinal data attributes in tabular and time-series formats (broad **E1**); and they provided candies in different colors, shapes, sizes, flavors, smells, and textures (broad **E2**). **E3** was moderately broad as they provided candies with various form factors, and **M1** was somewhat broad, as they used candy as the main workshop material but also provided craft materials such as wooden sticks, rubber bands, and paper plates. Their research questions did not focus on **M2**, **M3** or **I1**, which neither considered in the workshop design nor in its analysis.

★ Zizzi et al. [25] conducted a series of workshops with six dancers to explore the parameters of human body-movement that could be used to represent data. They kept **M1**, **E1**, and **E3** specific. For **M1**, the material is the human body (we consider movement as **E3**: physical mark). They also kept **E1** specific, as they worked only with quantitative data. As they studied various body-movement techniques (e.g., ballet, contemporary, Yoga), they kept **M2** somewhat broad (but still limited). They progressively expanded their initial set of existing human body-movement variables after each iteration, making **E2** broad. Due to the importance they placed in somatic experiences when representing data, they kept **I2** broad.

5 GENERATIVE POWER OF FLEXPHYS

In addition to descriptive and comparative power, useful models have generative power [2]. In this section, we show how FlexPhys can be used to generate three example workshop scenarios that illustrate a range of operationalizations. We discuss each scenario by providing the type of research question (**RQ**) (general questions that are often asked in visualization research), the specific instantiation of that research question (**SQ**), and a possible operationalization using FlexPhys. For conciseness, we focus on considerations that are not necessarily straightforward at first, but that are brought to light by considering the FlexPhys ingredients.

RQ1: How might we represent data with a specific variable?

SQ1: How might we encode nominal, ordinal, and quantitative data into the tactile variables ‘roughness’ and ‘texture’?

Operationalization: The researchers assume that the main goal of the physicalizations would be to convey information so that it would be perceived accurately, for example in a context with blind or low-vision users (context and goal). To investigate different ways of representing nominal, ordinal, and quantitative data (broad range of data types **E1**) into roughness and texture (specific variable **E2**), researchers might think of a workshop in which participants encode data into the texture and roughness of various samples created using 3D prints (similar to an existing study on encoding data into roughness [8]). However, FlexPhys highlights additional aspects that could be overlooked but should be considered. The properties of different materials (**M1**) is one such aspect. We know that texture in different materials such as fabric, wood, and plastic are perceived differently and is associated with different emotions [13]. Therefore, the researchers might decide to provide a range of materials with different textures and roughness (e.g., solid materials like glass, sanded wood and stone, and flexible or malleable materials like fabric and clay). To narrow down the scope of the study and because different techniques and tools exist for each of these materials, the researchers could decide to provide tokens of each material as marks (**E3**) and decide not to consider material-specific techniques (**M2**) or tools (**M3**). They might also decide to keep the sensory modality (**I1**) and interaction (**I2**) open so that workshop participants can consider the different ways in which the texture- and roughness-based physicalizations would be perceived and interacted with (e.g., through sight, touch, or sound).

RQ2: How might we represent data with a specific material and/or technique?

SQ2: How might we represent data in fabric and what are the properties of fabric that can be used for data representation?

Operationalization: Their narrow focus on fabric as material (**M1**) allows them to study fabrics with wide ranges of properties (e.g., wide ranges of elasticity, absorbency, strength, breathability, drape and weight). These wide ranges of properties foster thinking about the variables that fabric affords for data representation (**E2**). In considering **M2**, the researchers find that several textile techniques have potential for representing data [1, 4, 19, 15]. They decide to provide a broad range of techniques to work with textile, and to support these techniques by providing a broad range of tools (**M3**) that

participants can use for cutting, sewing, and stitching for example. Continuing their systematic consideration of FlexPhys, they need to consider the kind of data they will provide. This makes them realize that encoding needs and preferences will likely be influenced by the type of data to be represented. Therefore, they decide to provide a few datasets about different topics but all including some nominal, some ordinal, and some quantitative data dimensions (**E1**). By doing so, they aim to gather information about which data types might be more suitable to data physicalization with fabric. They further decide to keep the mark open (**E3**), so participants can define the units to represent data, and to keep interaction open-ended (**I1**) so that participants can explore how data will be perceived and which sensory modality (**I2**) will be used. Participants can also define the context and goal of the physicalization based on their personal preference and the dataset they choose (context and goal).

RQ3: How might we represent data physically for a specific application and context to achieve a specific goal?

SQ3: How might we represent carbon footprint data in urban space design to improve social awareness?

Operationalization: In the context of urban space design and with the goal of enhancing awareness about climate action and carbon footprint among citizens, researchers aim to conduct workshops with urban designers and people with a background in visualization (context and goal). The key consideration for them is to promote interaction through participatory design methods (**I1**) for increased engagement—as this is a requirement from their community partners. To maximize the chances that participants generate many ideas, they use FlexPhys to broaden the scope of the workshop and proactively avoid making decisions that might restrain their participants’ creativity. They decide to keep the materials, techniques, and tools broad (**M1**, **M2**, **M3**). They also provide meaningful datasets about carbon footprint where the data type is not an important aspect, as long as these are realistic datasets (**E1**). They also keep variables and marks open-ended (**E2**, **E3**) so that participants can explore the possible elements of the urban environment with the potential to represent data and their properties that can act as variables for data representations. They decide to keep sensory modality open to encourage multisensory representations (**I2**) that may foster community engagement.

6 FUTURE WORK AND CONCLUSION

FlexPhys presents a starting point for future research and evaluation of data physicalization workshop designs. We provide this cookbook as a tool for systematically exploring different variations of the workshops. Future researchers can use the cookbook to address questions about which ingredients to use, their impact on the workshop design and expected outcomes, in regard to the research questions they want to answer.

We also envision FlexPhys to evolve as new research questions are investigated through workshops. This initial cookbook is informed by the analysis of four specific workshops, and therefore not providing an exhaustive list of possible ingredients. In fact, we expect that researchers designing and running data physicalization workshops in the future will identify additional ingredients. Those ingredients might directly fit into the FlexPhys cookbook, but it is also possible that some dependencies, or even conflicts, emerge from those additions. Therefore, we do not argue that FlexPhys should be used ‘as-is’ or set in stone, but rather seen as an evolving tool that will help researchers design, think through, and document similar workshops.

In FlexPhys we focused on workshops that involve some form of hands-on data physicalization making. However, there are other kinds of physicalization workshops based on sketching, card-based activities, or simply discussion among participants. Future research should look into expanding the cookbook to support other kinds of data physicalization workshops.

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SUPPLEMENTAL MATERIALS

The FlexPhys cookbook template and an annotated cookbook are made available for reference in supplemental materials, at <https://hal.science/hal-05151375>.

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